

Review

Recent Advances in Energy Chemistry between Solid-State Electrolyte and Safe Lithium-Metal Anodes

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Because of the high capacity of lithium (Li) metal and the intrinsic safety of solid-state electrolyte, solid-state Li-metal batteries are regarded as a promising candidate for next-generation energy storage. However, uncontrollable dendrite growth and large interfacial resistance severely hamper the practical applications. This review summarizes the issues generated by the marriage of Li-metal anodes and solid-state electrolytes. First, the current challenges are underscored. Specific attention is paid to the large interfacial resistance, uncontrolled dendrite growth, and low operation current or capacity. The second section is dedicated primarily to understanding the ionic channels in the composite electrolyte and the space charge layers in the interfacial region. Based on these dilemmas and working principles, emerging strategies to render solid-state Li-metal batteries are summarized. Finally, the general conclusion and perspective on the current limitations and recommended research of solid-state Li-metal batteries are presented.

INTRODUCTION

Energy is the everlasting impetus for a sustainable society. Among all current energy conversion and storage systems, the battery is one of the most convenient devices to effectively utilize energy. The commercialization of rechargeable lithium-ion batteries (LIBs) in the early 1990s is considered the most significant milestone in the development of electrochemical energy storage technologies. Since then, LIBs in portable electronics and electric vehicles have greatly innovated our modern lifestyle. With the continuous upsurge in demand for energy storage, future batteries will concentrate on further improving the energy density and achieving an adequate safety.¹

Exploring advanced electrode materials with a high capacity and a large voltage window is the primary strategy for achieving a cell with a high energy density. Among various electrode materials, Li metal renders an ultra-high theoretical specific capacity (3,860 mAh g⁻¹, 10 times higher than that of graphite anode in LIBs) and the most negative electrochemical potential (−3.040 V versus standard hydrogen electrode).² When Li-metal anode matches a cathode, the as-obtained cell can potentially realize a higher capacity and a larger operating voltage, thus achieving a higher energy density. Therefore, Li-metal batteries (LMBs) can promisingly power the electronic products for a longer time while remaining lightweight.³

In conventional cells, non-aqueous electrolytes are frequently adopted to transport Li ions between the cathode and anode.⁴ The flammability and easy-leak feature of non-aqueous electrolyte induce a severe safety concern. The replacement of a non-aqueous electrolyte with a safe solid-state electrolyte (SSE) presents a promising

The Bigger Picture

Li metal is extensively focused because of its ultra-high theoretical capacity (3,860 mAh g⁻¹, 10 times higher than that of graphite) and the most negative electrochemical potential (−3.040 V versus standard hydrogen electrode). Relative to non-aqueous electrolyte, solid-state electrolyte exhibits enhanced safety. Therefore, the marriage of Li metal and solid-state electrolyte is expected to inherit the merits of both. However, the competitive solid-state Li-metal batteries encounter formidable challenges, including dendrite issues and poor solid-solid contact. Investigating the interdisciplinary issues of solid-state electrolyte and Li metal in integrated cells is necessary to realize efficient and safe solid-state Li-metal batteries. This review focuses on the energy chemistry issues generated from the marriage of Li metal and solid-state electrolyte instead of their respective problems. The current strategies, limitations, and emerging research directions are presented.

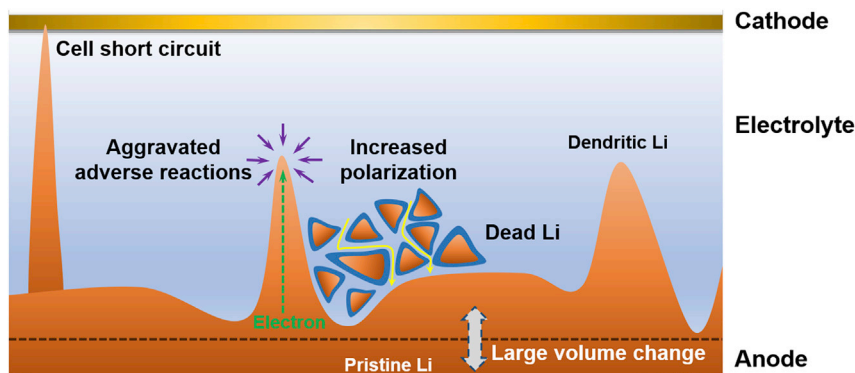


Figure 1. Dilemmas of Li-Metal Anodes

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route to enhancing the safety performance of cells. Additionally, SSE is convenient to achieve an ultrathin thickness, which can hopefully improve the energy density and apply to flexible and wearable devices.⁵ Therefore, when Li-metal anode meets SSE, solid-state LMBs (SSLMBs) with SSE and Li-metal anode can potentially deliver a high energy density and excellent safety assurance.^{6,7} SSLMBs are on the cusp of being a commercially available technology.

However, the fantastic SSLMBs encounter formidable challenges on the way to practical applications.⁸ (1) Li-metal anode was proposed in the 1960s, but it is long forsaken in commercial devices because of dendrite growth, infinite volume change, and severe chemical and electrochemical reactivity of Li metal. These issues result in serious side reactions, safety concerns, and poor lifespan of LMBs (Figure 1).⁹ The advances in electrolyte and electrode design facilitated by nanoscience and technology in the current century afford an emerging opportunity to revive Li-metal anode.

These problems and methods have been clearly explained by our previous reviews^{9–11} and other literature,^{12–16} which will not be repeatedly introduced herein. (2) SSE, with high expectations to enhance the safety and energy density of batteries, still confronts several troubles, such as low ionic conductivity ($10^{-8} \sim 10^{-3}$ versus $10^{-4} \sim 10^{-2} \text{ S cm}^{-1}$ for routine non-aqueous electrolyte), inferior solid-solid contact with electrode materials, and poor maneuverability as a result of its fragile nature. A large polarization is induced by the sluggish Li-ion diffusion through the bulk electrolyte and electrode-electrolyte interface. Hence, solid-state batteries have to be operated at a higher temperature to improve the ionic conductivity and promote the solid-solid contacts. The stability,¹⁷ ion transport,¹⁸ and failure mechanism¹⁹ of sulfide SSE^{20,21} and garnet SSE^{22–26} have been illustrated in previous literature. These reviews summarize the significant progresses of Li-metal anode and SSE, which are greatly beneficial to achieving an efficient Li-metal anode or SSE.

However, when Li-metal anode matches SSE, their respective issues become more complicated, and new troubles become apparent: (1) Relative to the conventional Li-metal anode with non-aqueous electrolyte, Li-metal anode with SSE encounters emerging problems, such as the heterogeneous solid-solid contact between SSE and Li metal. This poor contact can result in a large interfacial impedance and short lifespan. (2) Compared with the conventional solid-state batteries with graphite anode, Li-metal anode with SSE has a severe dendritic Li deposition, which can lead to safety concerns and low efficiency. Additionally, contrary to the prevailing

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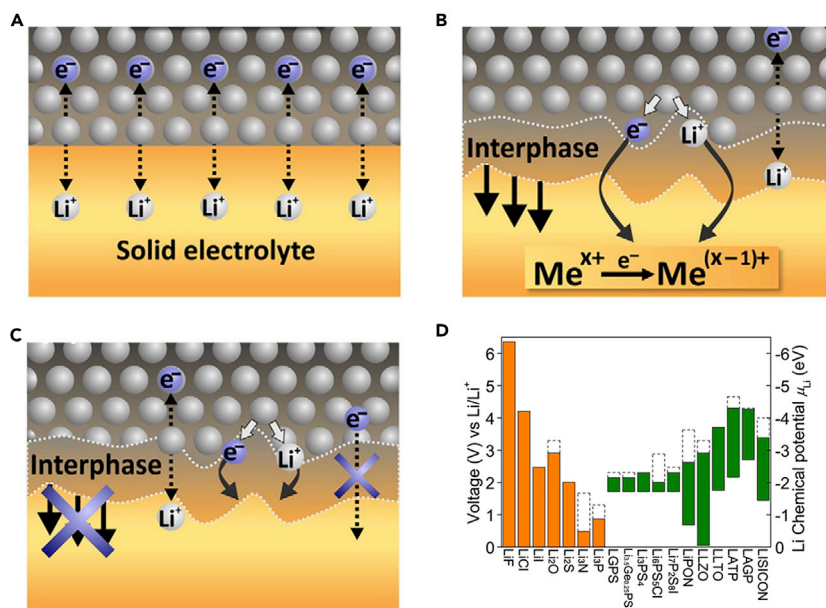


Figure 2. Interfacial Stability between SSE and Li-Metal Anodes

(A–C) Stable interface without any reactions (A), unstable interface with reactions to form mixed conducting interphase (B), and unstable interface with reactions to form SEI featured ion-conducting and electron-insulating (C). Reproduced from Wenzel et al.²⁸ Copyright 2015 Elsevier. (D) Stable voltage windows (versus Li/Li⁺) of some SSEs. Reproduced with permission from Zhu et al.²⁹ Copyright 2015 American Chemical Society.

theory that electrolytes with high mechanical strength can effectively suppress Li dendrite growth, the dendrite issue of Li-metal anode cannot be simply handled by SSE. Therefore, comprehensively investigating the interdisciplinary energy chemistry of SSE and Li-metal anode in an integrated cell is indispensable for constructing an efficient and safe SSLMB.

In this review, we summarize the energy chemistry issues generated from the marriage of Li-metal anode and SSE rather than emphasize their respective troubles. First, we present the multiple dilemmas of SSLMBs. Poor interfacial stability, dendrite growth, and low operation current and capacity of SSLMBs are underscored. Specific attention is paid to the basic principles of SSE-matching Li-metal anode, such as the ionic channels in the bulk SSE and the space charge layer in the interface. On the basis of these primary understandings and related analysis, recently proposed strategies to render an efficient and safe SSLMB are described. A general conclusion and a perspective on the current limitations and recommended future research directions of SSLMB are presented.

CURRENT DILEMMAS

Interfacial Stability

A perfect interface between SSE and Li-metal anode should be highly ionically conductive and electronically insulating,²⁷ which can freely transfer Li ions and avoid the side reactions between Li-metal anodes and SSE (Figure 2A).²⁸ However, because of the ultra-low electrochemical potential of Li metal, most SSEs cannot maintain stable when matching Li-metal anode with a transition layer formed (Figure 2D).^{29,30} (1) If the formed interface is a mixed-conducting layer (Figure 2B), which can still conduct electrons, the reactions between Li-metal anode and SSE will

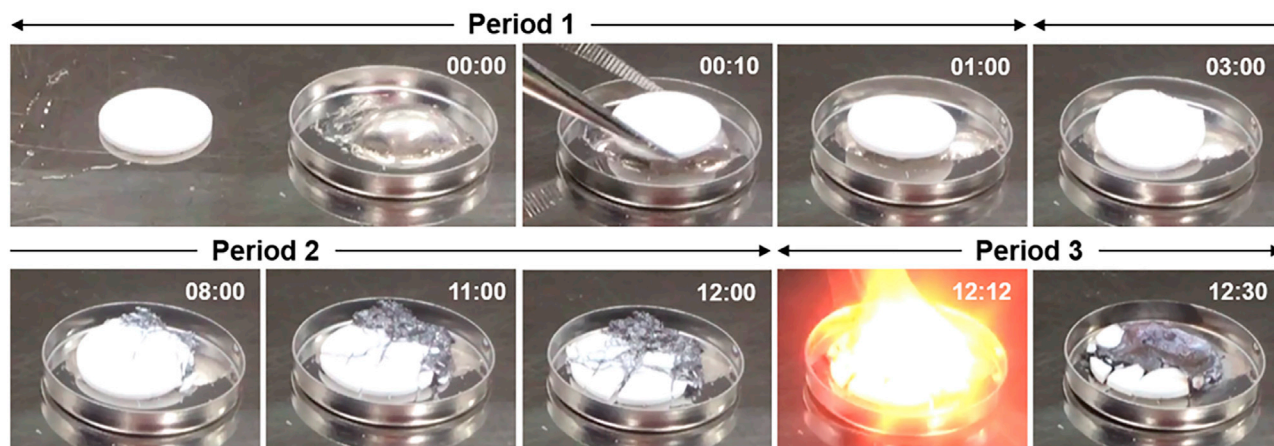


Figure 3. Thermal Runaway from the Reactions of $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ Pellet and Molten Li
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continually occur. The interface will further grow and finally result in cell failure. (2) If the formed interface is a metastable solid-electrolyte interphase (SEI), the electron transfer is therefore stopped (Figure 2C) and the as-formed interface can maintain stable toward Li-metal anode and SSE.³¹ Therefore, the reactions between Li metal and SSE are acceptable under the condition that the reaction product is ionically conductive and electronically insulating to stop the further reactions. The reactions between them can sometimes render a superior interface for evaluating the interfacial impedance, which is discussed in the following section.

Oxides such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) are regarded as relatively stable SSEs toward Li-metal anodes, while many other SSEs can chemically or electrochemically react with Li-metal anode. In the case of some highly conductive sulfide SSE, such as $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$, and $\text{Li}_{10}\text{Si}_{0.3}\text{Sn}_{0.7}\text{P}_2\text{S}_{12}$, the interfacial film formed from the interfacial reactions is a mixed ion-electron conductor.³² Electron transfer across this mixed conducting film causes a continuous decomposition of the solid electrolyte and a continuous growth of the interfacial film, which severely increases the diffusion resistance of Li ions in the long-term cycling. The continuous interfacial reactions can be effectively relieved by interfacial coating³³ and element doping into the SSE.³⁴

The unstable interface between Li metal and SSE can not only lead to poor Li-ion diffusion and a short lifespan but also cause potential safety concerns.³⁵ Chung et al. put a sintered $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ pellet in contact with molten Li metal at 200°C in a glove box with water and oxygen content below 0.1 ppm, and a drastic reaction between SSE and molten Li was observed.³⁶ A rapid kinetic reaction at high temperatures causes swelling of the interface with rapid emission of O_2 gas, and the interface goes through rigorous thermal runaway with a large flash (Figure 3, period 3). The continuous reactions are contributed by the mixed conducting interphases. The strong thermal runaway induces a safety hazard for the SSLMBs. Therefore, in some extreme conditions (such as thermal runaway), oxygen in the SSE can release and accelerate the possible burn and explosion. During the design of SSE, the stability of oxygen in the SSE and its effect on the cell safety should be further investigated.

Apart from the adverse aspects mentioned above, the products generated from the interfacial reactions between Li metal and SSE are reported to be able to wet the

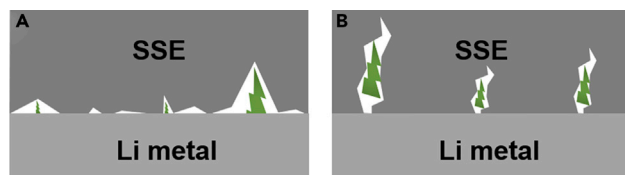


Figure 4. Dendrite Growth in SSE

(A) Dendrite growth occurs in the interface between Li-metal anode and SSE.

(B) Dendrite growth from Li-metal anode to the interior of SSE.

interface and suppress the dendrite growth. Wu et al. discovered that although $\text{Li}_{6.1}\text{Ga}_{0.3}\text{La}_3\text{Zr}_2\text{O}_{12}$ is stable against Li metal, Li dendrites can still form along the holes, phase boundaries, and defective sites of the $\text{Li}_{6.1}\text{Ga}_{0.3}\text{La}_3\text{Zr}_2\text{O}_{12}$ pellet.³⁷ In contrast, the reactions between $\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{P}_2\text{O}_5-\text{TiO}_2-\text{GeO}_2$ and Li effectively terminate the continuous Li dendrite propagation and growth. Therefore, the metastable interface can afford a soft contact between Li-metal anode and SSE and thus impede the continuous Li dendrite growth. On the contrary, a stable and rigid interface (such as that between $\text{Li}_{6.1}\text{Ga}_{0.3}\text{La}_3\text{Zr}_2\text{O}_{12}$ and Li metal) will terminate the cell cycling earlier as a result of the increased interfacial impedance and dendrite growth. Therefore, controlled reactions between Li metal and SSE are acceptable and sometimes favorable, and an SEI rather than a mixed conducting interface is required between Li metal and SSE.

Dendrite Growth

The prevailing knowledge is that SSEs have a strong ability to suppress dendrite growth because of their high mechanical modulus.³⁸ However, practically, Li dendrite growth is frequently observed when SSE is adopted for Li-metal anode, even at a relatively low current density of 0.5 mA cm^{-2} .³⁹

Dendrite growth with SSE is attributed to several factors:

- (1) The poor contact between SSE and Li-metal anode (Figure 4A). Holes generated from the non-conformal solid-solid contact afford the free room for dendrite growth. Dendrites can also grow into the interior of SSE in the following cases (Figure 4B).
- (2) Low compaction density of SSE. The technologies adopted during SSE production determine the compaction density of SSE. The low compaction density can also induce many holes in the SSE and result in dendrite growth.
- (3) Conductivity difference in the grain and grain boundary. Some SSEs have a high Li-ion diffusion rate in the grain boundary, resulting in the preferential deposition and dissolution of dendrites in the sites of grain boundary.⁴⁰
- (4) The soft phase in composite electrolytes. Polymer is typically adopted to work with inorganic SSE to improve the flexibility of composite SSEs. Relative to the mechanically strong inorganic SSE, dendrites are easier to grow in the soft polymer. Therefore, Li dendrite growth is still a formidable challenge in SSE.

Dendrite growth can result in a low utilization of the Li-metal electrode. Although the side reactions between SSE and Li-metal anode lead to the low utilization of Li metal, dendrite growth and its induced dead Li largely contribute to Li loss as well. Take the stable LLZO as an example: although no side reactions are believed to occur between LLZO and Li metal, dendrites still grow in the pores and grain boundaries of LLZO. During the discharging process, Li stripping is not uniform and the roots on the conductive matrix easily exchange electrons, making the top Li in the

heterogeneous interface and the LLZO a dead Li. Therefore, the low utilization of LLZO is not caused by the side reactions between Li metal and LLZO but by dendrite growth and the heterogeneous stripping process of deposited Li.

Dendrite growth can be relieved by modifications in the SSE and its interface.⁴¹ Han et al. incorporated Lil into $\text{Li}_2\text{S-P}_2\text{S}_5$ glass into the SSE to achieve high ionic conductivity and electronic insulation in the interface between SSE and Li-metal anode, thus alleviating the dendrite growth at the interface.⁴² However, it is a great challenge to suppress dendrite growth in the SSE. Elaborately designing the structure of the SSE and its interface is a must to minimize dendrite growth in SSLMBs.

Low Operation Current and Capacity

As a result of the interfacial instability and dendrite growth described above, both bulk and interfacial impedances of SSLMBs are very large, even when the bulk conductivity of some SSE such as sulfide electrolytes almost reaches that of the non-aqueous electrolytes ($10^{-2} \text{ S cm}^{-1}$). Therefore, the Li-metal anode with SSE cannot work at a large current and capacity. The operation current ($0.01 \sim 3 \text{ mA cm}^{-2}$) and capacity ($0.0025 \sim 3 \text{ mAh cm}^{-2}$) for SSLMB is still ultra-low in relation to that in the non-aqueous electrolytes ($0.1 \sim 10 \text{ mA cm}^{-2}$, $0.5 \sim 10 \text{ mAh cm}^{-2}$) in the practical cells.⁹ Therefore, contrary to a widespread consensus, routine SSE cannot perfectly cope with the dilemma of Li-metal anode. Constructing interconnected ion-conductive matrix within the cathode, anode, and separator is a feasible solution to reduce the interfacial impedance. A greater mechanistic understanding and novel strategies are critically required to solve the dendrite and interfacial issues of SSLMBs.

BASIC PRINCIPLE

Ionic Channels

Current SSEs can be divided into two categories: polymer and inorganic ceramic. The polymer electrolyte exhibits better flexibility and thus better interfacial contact with Li. However, the relatively low ionic conductivity and narrow electrochemical window limit its application in a high-voltage battery. Ceramic electrolyte usually possess a desirably high modulus for dendrite suppression and a higher ionic conductivity. However, most inorganic ceramics are very fragile, which restricts their application in mass production. Therefore, polymer or ceramic composites are strongly regarded as the potential SSE for practical applications. Different from the composite polymer electrolytes containing inert fillers such as Al_2O_3 , TiO_2 , or SiO_2 , the composite electrolytes herein contain rich ion-conductive ceramics (ceramic % > 50 wt %). An increased ionic conductivity and flexibility are combined for composite electrolytes because of the high ionic conductivity of the ceramic accompanied by the reduction in crystallinity of the polymer electrolyte and the binding role of polymer electrolyte.

It is primarily important to investigate the ionic channels in the composite SSE. The following three channels can be responsible for the ionic conduction in a polymer or conductive ceramic composite electrolyte:

- (1) Conductive ceramic: Relative to the polymer electrolyte, conductive ceramic has a high ionic conductivity. In the composite electrolytes with the content of ceramic exceeding percolation threshold, Li ions tend to go through conductive ceramic. By combining selective isotope labeling and high-resolution solid-state Li NMR, Li-ion pathways within LLZO and polyethylene oxide (PEO) composite electrolytes are tracked through monitoring the

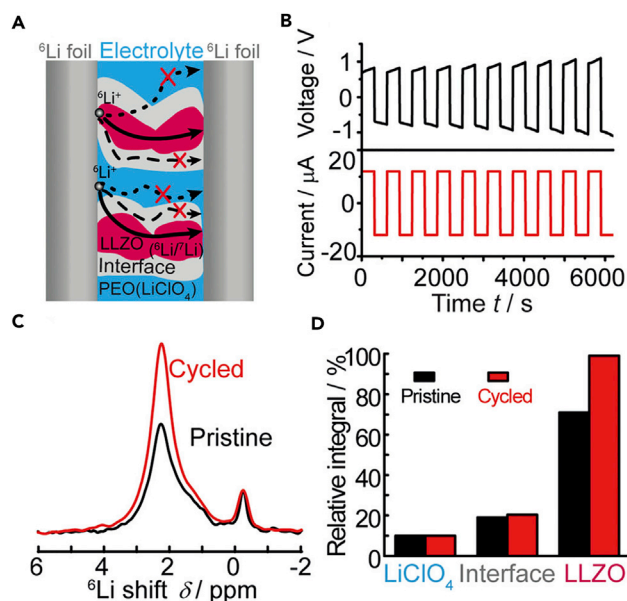


Figure 5. Ionic Channels in the Composite SSE

(A) Schematic diagram of the symmetric ^6Li foil/composite SSE/ ^6Li cell and possible Li-ion transport paths within the composite SSE.

(B) Electrochemical profile of ^6Li foil/composite SSE/ ^6Li cell cycled with a constant current and plating and stripping time of 300 s.

(C) Comparison of the ^6Li NMR spectra of LLZO-PEO (LiClO_4) composite SSE before (pristine) and after (cycled) cycling.

(D) Quantitative analysis of ^6Li amount in LiClO_4 , interface, and LLZO of the composite SSE before (pristine) and after (cycled) cycling.

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replacement of ^7Li in the composite electrolyte by ^6Li from the ^6Li -metal electrodes during battery cycling.⁴³ The results experimentally confirm that Li ions favor the pathway through the LLZO ceramic phase instead of the PEO-LLZO interface or PEO (Figure 5).

- (2) The interface between polymer and inorganic electrolyte: When Li ions migrate from inorganic ceramic phase to polymer phase, Li vacancy forms in the ceramic. The difference in Li concentration between two electrolytes results in the formation of a space charge layer, which also acts as a channel for Li-ion diffusion. This effect is strongly confirmed by the increased conductivity in the nanosized composite electrolyte, which has many interfaces and Li-ion diffusion channels. Nevertheless, Yu et al. discovered that the ion conduction within the grain boundary is related to the operation temperature.⁴⁴ Li-ion diffusion kinetics is comparable to the bulk in grain boundary at high temperatures (>900 K). In contrast, diffusion rate in the grain boundary remains 50% of the bulk at room temperature.
- (3) Polymer electrolyte: Due to its low conductivity relative to ceramic, ionic channels within polymer electrolyte in the composite SSE account for a relatively small proportion.

Therefore, the ionic channel in the conductive ceramic contributes greatly to the ionic conductivity of ceramic-rich composite electrolytes. Improving the content of ceramic and its bulk conductivity is superior to increasing the Li-ion transfer channels within the composite electrolyte. Nevertheless, the contribution of the ionic channel at the interface of ceramic and polymer should not be ignored because of the

Equilibrium (Atomic structure)

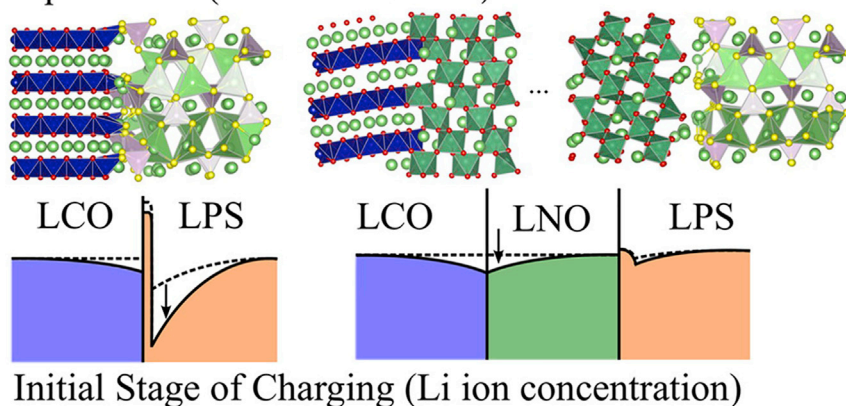


Figure 6. Space Charge Layer Effect between Oxide Cathode and Sulfide SSE

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complicated transfer behavior of Li ions in the composite electrolyte. Clearly, investigating the Li-ion diffusion paths in the composite electrolyte and rationally designing the composite structure will be beneficial in improving the ionic conductivity of the composite electrolyte.

Space Charge Layer

The poor solid-solid contact between SSE and Li metal leads to large polarizations and deteriorated cycling performance. Additionally, the uneven Li distribution inside SSE caused by the space charge layer makes the issues worse. These collectively contribute to the large resistance and low operation current and capacity of SSLMBs.

The space charge layer is caused by changes in the concentrations of charge carriers.⁴⁵ The space charge layer was first proposed by Wagner to explain the conducting effect of the interface of two semiconductors.⁴⁶ Then, Maier conducted comprehensive research on the space charge layer in many interfaces.⁴⁷ Haruyama et al. observed the space charge layer effect in the interface of SSE (β - Li_3PS_4) and oxide cathode (LiCoO_2).⁴⁸ By density functional theory (DFT) + U calculations, coupled with a systematic procedure for interface matching, the routine cathode interface (β - $\text{Li}_3\text{PS}_4/\text{LiCoO}_2$) exhibits attractive Li adsorption sites and rather disordered structure (Figure 6). Subsurface Li ions in the β - Li_3PS_4 side start to transfer at the undervoltage condition, resulting in the growth of the space charge layer and interfacial resistance. When LiNbO_3 buffer layers are intercalated between LiCoO_2 and Li_3PS_4 , smooth interfaces without Li adsorption sites for both LiCoO_2 and Li_3PS_4 sides are formed. The LiNbO_3 layer suppresses the space charge layer effect and provides smooth Li transport paths without penetrating through obstacles. Therefore, the space charge layer effect plays a significant role in cell cycling. Modification of the space charge layer at the atomic scale will further improve the performances of SSLMB.

The concept of space charge layer effect is also employed in the non-aqueous electrolyte, which is adopted to describe the anion depletion and dendrite growth.^{49,50} However, the mobility of anions in the SSE is negligible in comparison with that of liquid electrolytes and the space charge layer in the non-aqueous electrolyte is different from that in SSE.⁵¹ The nearly negligible electrochemical and chemical reactions between the Li-metal anode and LLZO reveal that there might be no

transition layer and the existence of a space charge layer between them is still unverifiable. Hence, it is difficult to clearly detect the space charge layer between the SSE and Li-metal anode, as there is no clear example of such an endeavor. Combinations of scanning probe microscope, electron holography, and DFT calculations can afford more information to identify the space charge layer between the SSE and Li-metal anode. Relieving the space charge layer effect is beneficial in reducing the interfacial impedance and achieving a superior cell-cycling performance.

As discussed above, there are still many dilemmas for SSE and Li-metal anode in the SSLMBs. Large interfacial and bulk resistance and dendrite growth result in low operation current and capacity in practical SSLMBs. Novel strategies are critically required. Though several fundamental issues of SSLMB, such as the ion conduction and space charge layer, are still not clear to researchers, several strategies prove effective at improving the cycling performance of SSLMBs. In the next section, we summarize the proposed strategies to present some insights into the design of efficient SSLMBs.

STRATEGIES AND METHODS

Composite Electrolyte

As explained in the previous section, composite electrolyte is promising for achieving high conductivity and proper flexibility in a practical SSLMB. Because of the high ionic conductivity and thermal stability of PEO electrolyte, a PEO-based polymer is frequently introduced to composite inorganic ceramic. PEO-in-ceramic electrolyte (80% ceramic),⁵² polyvinylidenefluoride-hexafluoropropylene/ $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$,^{53,54} $\text{Li}_7\text{P}_3\text{S}_{11}$ -PEO- LiClO_4 ,⁵⁵ $\text{Li}_{0.33}\text{La}_{0.557}\text{TiO}_3$ nanofiber-enhanced PEO,⁵⁶ and $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$ /PEO/succinonitrile⁵⁷ are proposed as novel composite electrolytes. The design of efficient composite electrolyte is aimed at wetting the interface and enhancing the ionic conductivity.

Wetting the Interface

In situ polymerization with conductive ceramic on the Li-metal surface is an important strategy for evening the interface.

Ju et al. fabricated a composite of poly(vinyl carbonate) and $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ SSE via *in situ* polymerization.⁵⁸ The *in situ* polymerization was realized via injection of 40 μL of vinyl carbonate- $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ solutions into a cellulose separator sandwiched between the cathode and Li-metal anode. After that, the cells were moved to an oven and thermally treated at 60°C for 120 hr to ensure successful polymerization. Finally, an intimate contact between SSE and Li-metal anode was realized.

Enhancing the Ionic Conductivity

For the inorganic ceramic, its conductivity strongly relates to its structure. Ceramic nanoparticles are conveniently adapted to the composite polymer electrolyte, while these nanoparticle fillers can largely compromise the high bulk conductivity of ceramics due to their random orientations.^{59,60} As indicated in the previous section, ion-conducting pathways within conductive ceramic-polymer composite electrolytes mainly exist in the ceramic particles and ceramic-polymer interfaces. Therefore, rationally designing the structure of ceramic and interfacial contact is beneficial for improving ionic conductivity.⁵⁷

Cui and co-workers reported a composite electrolyte with well-aligned $\text{Li}_{0.33}\text{La}_{0.557}\text{TiO}_3$ nanowires through lithography.⁶¹ Compared with randomly oriented nanowires, aligned nanowires render rapid ion conduction (Figure 7A). Liu et al. reported

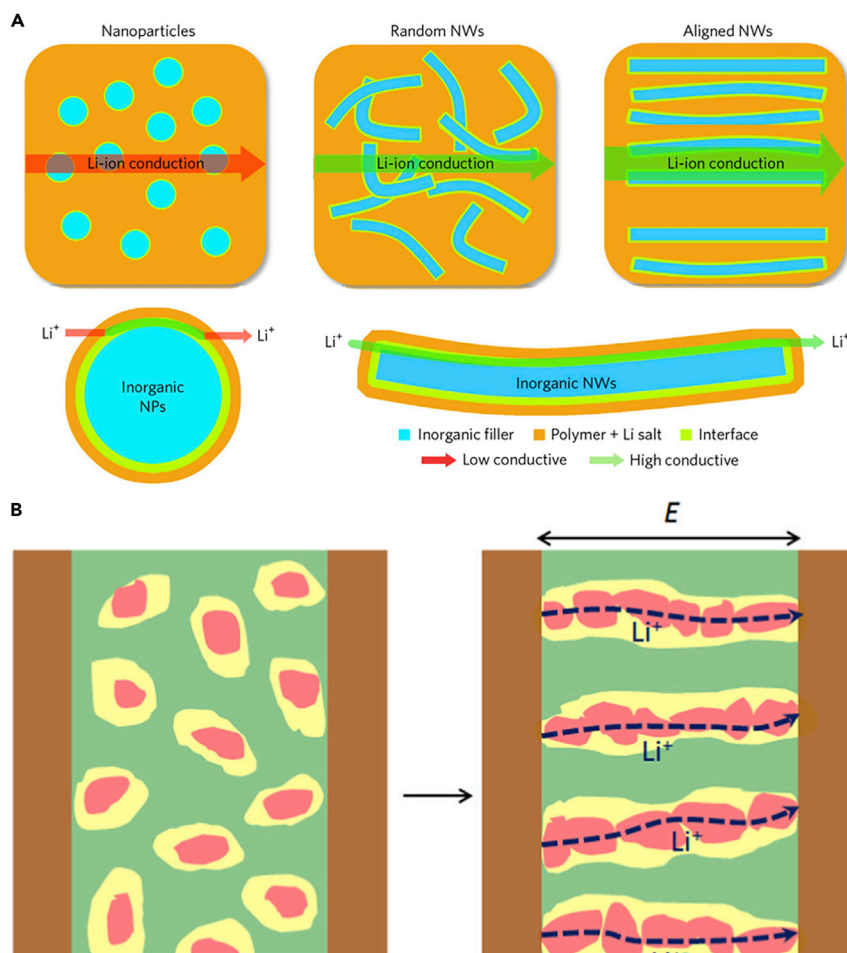


Figure 7. Aligned Nanowire in the Composite SSE to Enhance Ionic Conductivity

(A) Well-aligned $\text{Li}_{0.33}\text{La}_{0.557}\text{TiO}_3$ nanowires by means of lithography. Reproduced with permission from Liu et al.⁶¹ Copyright 2017 Springer Nature.

(B) The ceramic alignment by an external alternating-current electric field. Reproduced with permission from Liu et al.⁶² Copyright 2018 American Chemical Society.

a facile strategy for achieving the ceramic alignment through the dipole-dipole interaction by an external alternating-current electric field.⁶² The ceramic alignment realizes a high ionic conductivity at a high-voltage alternating-current electric field (Figure 7B). Through this strategy, the surface conductivity of the nanowires can be enhanced to a high value of $1.26 \times 10^{-2} \text{ S cm}^{-1}$ at 30°C , ten times the enhancement relative to the randomly dispersed nanowire-based composite electrolyte, nearly approaching the ion conductivity in liquid electrolyte. However, the composite electrolyte still cannot exhibit a high conductivity ($6.05 \times 10^{-5} \text{ S cm}^{-1}$ at 30°C) because of the limited amount of ceramic (<50 wt %). Further research can focus on improving the content of conductive ceramics to enhance the apparent conductivity of composite electrolyte.

Several composite SSEs have been proofed and demonstrated in special cell patterns. Principles can be further considered in the design of composite electrolyte for practical applications: (1) The primary role of the electrolyte is to conduct Li ions between the Li-metal anode and cathode. Hence, high ionic conductivity is the foremost target for a successful electrolyte. The composite electrolyte must

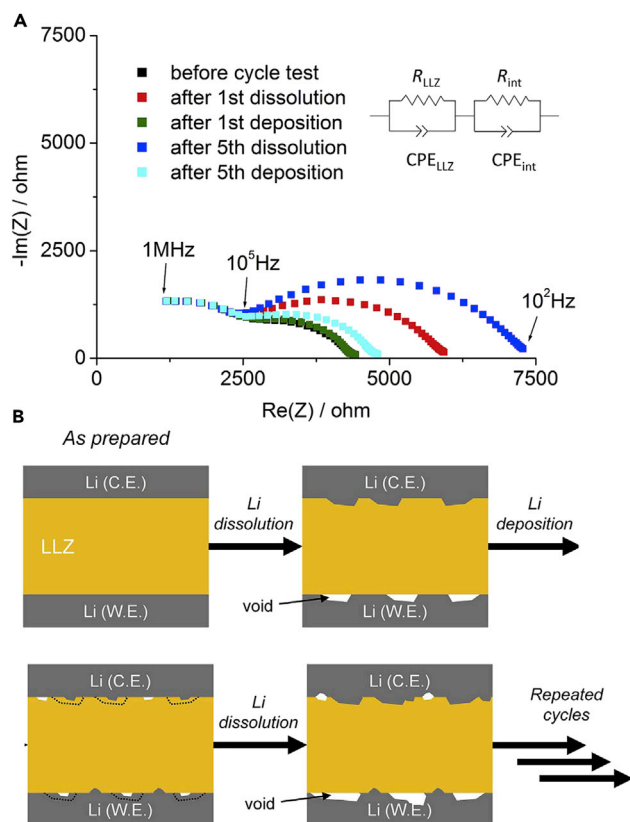


Figure 8. Interfacial Layer with Voids between SSE and Li Metal

(A) Impedance evolution during repeated Li plating and stripping.

(B) Schematic illustration of the dynamic changes at the Li-LLZ interface during repeated Li plating and stripping cycles.

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be rich in the highly conductive ceramic, while its flexibility can be realized by combining with the polymer electrolytes. (2) The composite electrolyte or its formed interface with Li metal must be stable against Li-metal anode without persistent reactions, which can reduce the interfacial impedance and render a long lifespan. On the basis of these considerations, more activities are required for designing and synthesizing the efficient composite electrolytes.

Interfacial Layer

Currently, the ionic conductivity of SSE can be enhanced to $10^{-3} \text{ mS cm}^{-1}$ at room temperature as a result of the extensive research performed on SSE.⁶³ If the conductivity of an SSE is 0.1 mS cm^{-1} and the thickness of the SSE is no more than $16.7 \text{ }\mu\text{m}$, the major voltage drop from the cathode to the anode is mainly derived from the large interfacial impedance between the SSE and the electrodes.⁶⁴ The poor solid-to-solid contact between the Li-metal anode and the SSE contributes primarily to the interfacial impedance. Koshikawa et al. investigated the dynamic changes that occur in the charge-transfer resistance at the Li- $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ interface during the Li plating and stripping cycles (Figure 8).⁶⁵ The resistance increases and decreases during Li stripping and plating, respectively, whereas the resistance does not go back to the initial value consistently after one cycle of Li stripping and plating. The increase and decrease in the interfacial resistances during Li stripping and plating, respectively, are most likely due to the formation and disappearance of voids at

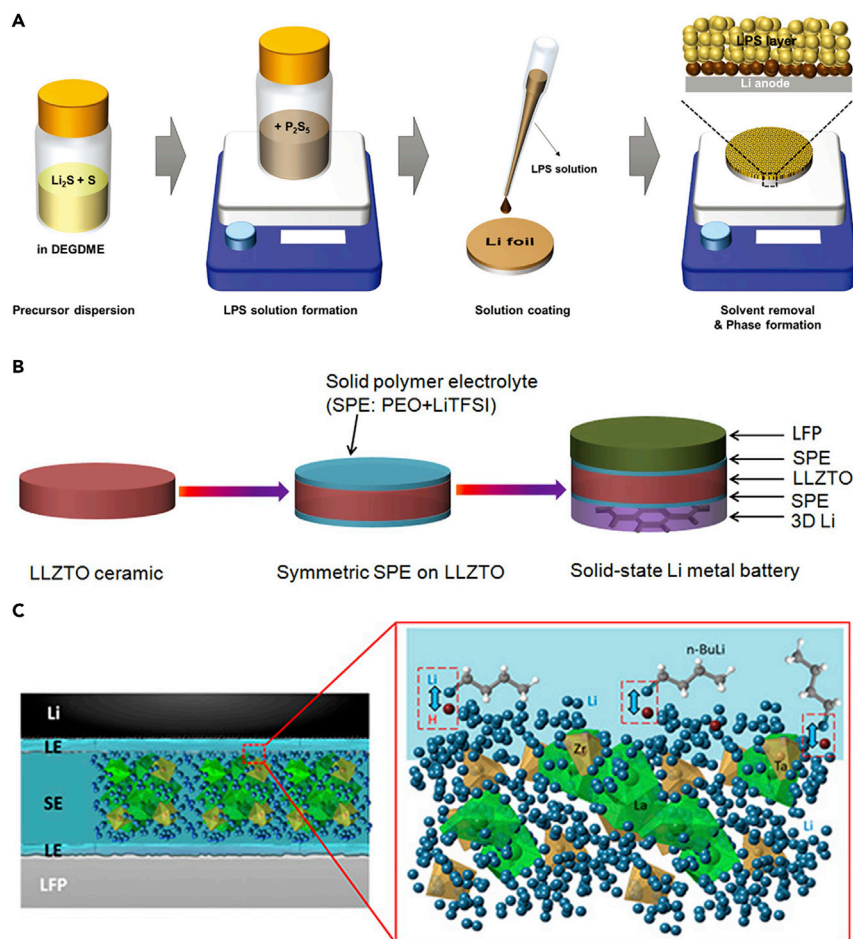


Figure 9. Interfacial Modifications between Li Metal and SSE

(A) *In-situ*-formed $\beta\text{-Li}_3\text{PS}_4$ SSE on Li-metal surface. Reproduced with permission from Lim et al.⁶⁶ Copyright 2018 Wiley-VCH Verlag GmbH & Co. KGaA.

(B) A soft polymer interface layer deposited on garnet electrolyte ($\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$) along with 3D Li-metal anode. Reproduced from Chi et al.⁸³ Copyright 2018 Elsevier.

(C) The non-aqueous electrolyte to wet the interface between Li metal and SSE. Reproduced with permission from Xu et al.⁸⁶ Copyright 2017 American Chemical Society.

the $\text{Li-Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ interface. The void formation during Li dissolution is suggested to be a critical factor that influences the interfacial resistance. Therefore, this interfacial impedance accounts for much of the voltage drop from the cathode to the anode for a highly conductive SSE.

For achieving an intimate contact between Li metal and SSE, drop-casting the SSE slurry onto the Li-metal anode surface is a facile method. Lim et al. prepared a $\beta\text{-Li}_3\text{PS}_4$ solution and directly drop-casted it on Li metal (Figure 9A).⁶⁶ The formation of an Li_3PS_4 SSE intimately coated on Li metal was achieved after slow evaporation of the solvent. Further, some SSE can be formed *in situ* on Li metal as a result of the reactions between Li metal and the precursor. Pang et al. adopted the reactions of Li, Li_2S_6 , and P_2S_5 to form Li_3PS_4 *in situ* on the Li-metal surface,⁶⁷ whereas Lu et al. utilized the reactions of Li, S, and P_2S_5 to form Li_3PS_4 *in situ* on the Li-metal surface.⁶⁸ The SSE formed *in situ* can form a tight contact with Li metal and maintains its chemical and electrochemical stability over repeated plating and stripping cycles.

Non-aqueous electrolyte reaches and wets the electrode surface automatically, whereas the solid (SSE)-solid (electrode) contact is really hard to be intimate. Interfacial modifications are frequently adopted to enhance the solid-solid contact between the SSE and Li-metal anode.⁶⁹ The modifications can be divided into three categories.

Alloy Layer

SSE-Li-metal contact is difficult to be intimate. The poor contact cannot be effectively relieved by high-temperature pretreatment due to the lithiophobic nature of many SSEs.³⁶ Hu and co-workers proposed an interface design between Li metal and SSE to improve the wettability, such as Mg coating to form the Li-Mg alloy⁷⁰ and ZnO.⁷¹ Al_4Li_9 ⁷² and Au⁷³ coating can also be introduced to wet the interface.

The above-mentioned interfacial layer is generally realized by a complicated method, such as atomic layer deposition. An easy and effective strategy was proposed by Shao et al. to enhance the interfacial connection between garnet SSEs and Li metal by drawing a graphite-based soft interface with a pencil.⁷⁴ When this graphite-coated $\text{Li}_{5.9}\text{Al}_{0.2}\text{La}_3\text{Zr}_{1.75}\text{W}_{0.25}\text{O}_{12}$ contacts molten Li, the graphite is lithiated to lithiophilic LiC_6 , achieving an intimate contact between Li metal and SSE. The *in situ* protection layer maintains chemical and electrochemical stability over repeated plating and stripping cycles.⁷⁵ After lithiophilic coating, the interfacial impedance can be reduced from several thousands to several ohms, rendering a long lifespan SSLMB.

Flexible Polymer

Flexible polymer can serve as a buffer layer between Li metal and SSE if the polymer itself remains stable toward Li metal. The protective buffer layer can circumvent the intrinsic chemical stability of SSE to Li metal, stop the migration of mixed ionic-electronic reactants to the inner of SSE, and improve the sluggish Li-ion diffusion kinetics at the interface by smoothening the interface. Guo and co-workers conducted a comprehensive work on the *in situ* polymerization of various polymers on Li-metal anode surface to realize a tight contact between Li metal and SSE, such as LiPF_6 -induced cationic polymerization of the non-aqueous electrolyte,⁷⁶ ultra-violet-curing,⁷⁷ and *in situ* plasticization with two conducting polymer networks.⁷⁸ Wang et al. employed PEO-500000 interfacial film to interconnect $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ -based electrolyte and an Li-metal anode.⁷⁹ The assembled Li-PEO/ $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$ -PEO/ $\text{LiMn}_{0.8}\text{Fe}_{0.2}\text{PO}_4$ all-solid-state cell exhibited an initial discharge capacity of 160.8 mAh g^{-1} with good cycling stability and rate performance at 50°C . Chinnam et al. realized the silanization at the SSE-Li-metal interface by $(\text{CH}_3\text{CH}_2\text{O})_3\text{Si}-(\text{OCH}_2\text{CH}_2)-\text{OCH}_3$ and LiTFSI,⁸⁰ which achieved a good adhesion and wetting and a low resistance between SSE and Li-metal anode, thus preventing the corrosion of the SSE ($\text{Li}_2\text{O}-\text{Al}_2\text{O}_3-\text{SiO}_2-\text{P}_2\text{O}_5-\text{TiO}_2-\text{GeO}_2$) by Li metal. However, most polymer electrolytes can react with the Li-metal anode, and an SEI rather than mixed conducting film must be formed to achieve a stable interfacial contact. A clear understanding of the chemical and mechanical properties of polymer coating layers on the Li-metal anode is critically important for designing and modifying these interfacial layers.⁸¹

When the flexible polymer interfacial layer matches the SSE and three-dimensional (3D) composite Li-metal anode, the cells are expected to achieve an elevated cycling performance and assurance of safety.⁸² Chi et al. demonstrated a soft polymer interface layer deposited on garnet electrolyte ($\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$) along with a 3D Li-metal anode (Figure 9B).⁸³ The interconnected interface between the SSE

and the electrodes is contributed by the polymer layers to settle the interface contact issue. 3D Li-metal anode can suppress dendrite growth by increasing the number of Li-ion deposition sites to reduce the local current density and render a uniform Li nucleation with 3D frameworks. With this ingenious design, the all-solid-state 3D Li | polymer-Li_{6.4}La₃Zr_{1.4}Ta_{0.6}O₁₂-polymer | 3D Li symmetrical cell exhibits a stable voltage profile over 700 h. This is a great breakthrough in the high-energy-density and safe LMBs. It will be of great significance if the cell can be operated at room temperature.

Non-aqueous Electrolyte Wetting

Because of the superior wetting performance and ability to form a stable SEI film,^{84,85} non-aqueous electrolyte is regarded as a potential candidate to wet the interfaces between Li-metal anode and SSE. Compared with the above-mentioned metal and oxide layer, non-aqueous electrolyte can efficiently wet the interface and render a great room-temperature cycling performance. To maintain the safety feature of solid-state working cells, the volume of non-aqueous electrolyte must be limited to a small content. Xu et al. added a small amount of liquid electrolyte including *n*-BuLi additive between SSE and Li metal to form a stable interfacial film (Figure 9C).⁸⁶ The interfacial stability is greatly enhanced with an almost constant SSE-Li-metal interface areal resistance during cycles. The hybrid LiFePO₄ | Li cell with *n*-BuLi between the SSE and the Li-metal anode can be stably operated for 400 cycles at room temperature at current densities of 100 and 200 $\mu\text{A cm}^{-2}$. The volume of the liquid electrolyte can be further reduced to 2 $\mu\text{L cm}^{-2}$ ⁸⁷ and 1.8 $\mu\text{L cm}^{-2}$,⁸⁸ which can still effectively decrease the interface impedance between the electrode and electrolyte.

Screening of proper non-aqueous electrolyte is of critical significance to achieve a superior interfacial layer. The liquid electrolyte reacts with the Li metal, while the products of the reactions should be stable toward Li metal. The as-formed interfaces between Li metal and SSE are expected to be a robust SEI rather than a mixed conducting interphase. Several recipes of non-aqueous electrolyte have exhibited critical roles in protecting Li-metal anode by forming an effective interfacial film, presenting fresh insights in designing these wetting agents. Gao et al. adopted a nanocomposite as an interface to protect Li₁₀GeP₂S₁₂ from Li metal.⁸⁹ The nanocomposite is constructed by the electrochemical decomposition of a liquid electrolyte (7.0 M LiTFSI in 1,3-dioxolane/1,2-dimethoxyethane) containing organic elastomeric salts (LiO-(CH₂O)_{*n*}-Li) and inorganic nanoparticle salts (LiF, -NSO₂-Li, and Li₂O). Stable Li electrodeposition over 3,000 hr and a 200-cycle life for a full cell were achieved.

To endow flexibility to Li_{6.75}La₃Zr_{1.75}Ta_{0.25}O₁₂ SSE, Zhao et al. coated Li_{6.75}La₃Zr_{1.75}Ta_{0.25}O₁₂ onto the polypropylene separator and added 7.0 $\mu\text{L cm}^{-2}$ electrolyte to wet the SSE-Li-metal interface.⁹⁰ With the decomposition of electrolytes during the charge-discharge process, the SEI layer forms on the SSE-polypropylene separator. The artificial SEI-SSE-polypropylene structure is eventually achieved as a working electrolyte, providing Li-ion-conducting properties but preventing electrical conduction. This SEI-SSE-polypropylene structure can also be realized by Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ SSE.⁹¹ The Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ particles are densely wrapped by uniform artificial SEI consisting of LiF, Li₂CO₃, and ROCO₂Li. This artificial SEI-Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃-polypropylene structure can effectively suppress Li dendrite after 450 cycles at 1C rate.

Non-aqueous electrolyte is frequently adopted to wet the interfaces, while most of the wetting processes are in a black box. Although many of these efforts render

superior cycling performance of the SSLMBs, the underlying mechanism remains unclear. Strictly speaking, a cell containing a liquid component is not all solid state, and its safety feature can vary depending on the liquid percentage. The goal is to keep the amount of wetting agents as low as possible to maximize the overall cell safety while maintaining its performance. To achieve this goal and rational design of the wetting agents, underlying mechanisms of non-aqueous electrolyte wetting should be better understood. Therefore, future research should focus on further reducing the volume of liquid electrolyte and exploring its functional mechanism in the solid-state cells.

These proposed strategies can effectively wet the interface and reduce the interfacial impedance between Li metal and SSE, which can render long stable cycles. In the future researches on the interfacial layer, three principles should be considered: (1) the interfacial layer should be thin and not increase the cell weight and potential drop very much, and (2) for the non-aqueous electrolyte wetting, reactions between non-aqueous electrolyte and Li metal should be accurately controlled to maintain the less non-aqueous electrolyte in the solid-state cells after the activation processes. The interfacial relations between non-aqueous electrolyte, Li metal, and SSE should be clearly characterized. (3) Similar to the non-aqueous electrolyte, the interface formed between SSE and Li metal should also suppress dendrite growth. Implanting the concepts and methods of the interface formation from non-aqueous electrolyte can shed some new light on the interfacial film construction in the SSE system. Through the regulation of the interfacial chemistry of the layers, a tight contact between the Li-metal anode and SSE can be achieved.

Mixed Conducting Framework

During cell charging and discharging, electrons and Li ions should both reach the electrodes to complete the electrochemical redox.⁹² Therefore, a mixed conducting framework that transports both electrons and Li ions is indispensable to allow the electrons and Li-ion diffusion to reach the electrodes. Li metal itself serves as a mixed conductor, but planar Li causes serious dendrite growth and insufficient contact with SSE. In liquid electrolyte, 3D Li-metal host is frequently adopted to relieve the severe volume change and dendrite growth problems. However, SSE cannot permeate into the host as easily as the liquid electrolyte does. Because electrochemical redox generally occurs at the interface between the SSE and the electrode, the inner part of the electrode framework is completely inert when there is no efficient electron and ion transport. Therefore, learning from the current cathodes in solid-state batteries where active materials, SSE and conductive agent are mixed to build electron and ion pathways, the anode of an SSLMB also requires a mixed electron-ion conducting framework.

Much effort has been devoted to the search for an effective mixed conducting framework. McDonald et al. developed a dual electron and Li-conducting polymeric composite with poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonic acid) and Li-conducting poly(ethylene oxide).⁹³ An ionic conductivity of $\sim 10^{-4}$ S cm⁻¹ and an electronic conductivity of ~ 45 S cm⁻¹ were achieved at an optimized ratio of polymer contents. However, it is difficult for most conductive agents to achieve high ionic and electronic conductivity by themselves at the same time. Mostly, electronic and ionic paths are provided by different agents.

The electronic pathways in the mixed conducting framework are generally contributed by the electron-conductive host. The conductive hosts reported so far can be

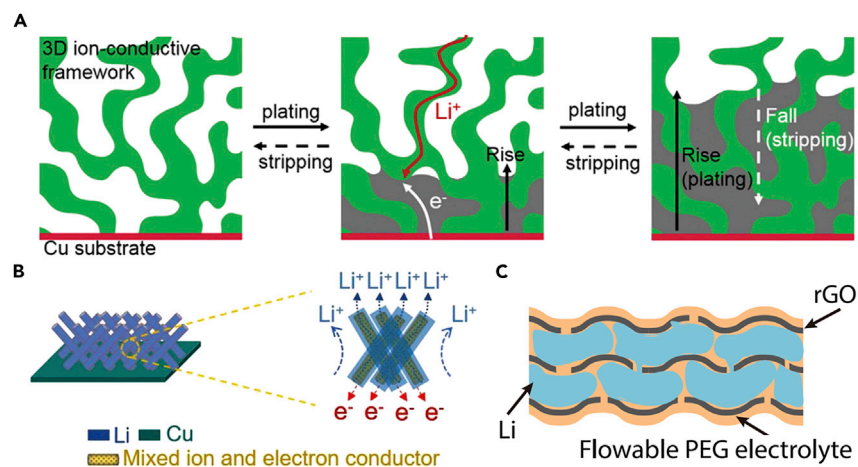
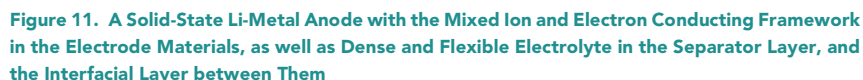


Figure 10. Mixed Conducting Framework in the Solid-State Li-Metal Anode

(A) 3D garnet-type ion-conductive framework as ionic paths and Li metal itself as the electronic paths. Reproduced from Yang et al.⁹⁸ with permission from the National Academy of Sciences.
 (B) Incorporating $\text{Li}_{6.4}\text{La}_3\text{Zr}_2\text{Al}_{0.2}\text{O}_{12}$ nanoparticles into 3D carbon nanofibers to fabricate a mixed ion- and electron-conducting network. Reproduced with permission from Zhang et al.⁹⁹ Copyright 2018 Wiley-VCH Verlag GmbH & Co. KGaA.
 (C) Poly(ethylene glycol) gel electrolyte as flowable interface into 3D Li-rGO. Reproduced from Liu et al.,¹⁰⁰ licensed by CC BY 4.0.

divided into two categories: nanostructured and microstructured materials. The nanostructured host renders a high specific surface area, which can largely reduce the areal current density and render a uniform deposition of Li metal.⁹⁴ The nano-scale platform is also convenient for achieving a uniform distribution of lithiophilic sites by element doping (such as nitrogen).⁹⁵ The microstructured host has a relatively large pore and matrix size at $1\sim 100\ \mu\text{m}$.⁹⁶ Li ions are more inclined to deposit in the pores, which can effectively relieve the volume change and dendrite growth issues.⁹⁷ Additionally, because of its conductive nature, Li metal itself can act as an electronically conductive agent, allowing electronically insulating materials to host Li metal. Inspired by this, Yang et al. constructed a 3D garnet-type ion-conductive framework as a host with ionic paths for the Li-metal anode (Figure 10A).⁹⁸ Li metal is plated from the bottom and rises during deposition within the solid ion-conductive host. The unique structure can render deposited Li free from electrolyte penetration and short circuit. However, there is no extra electronic path except Li in the matrix. If dead Li forms and break the electron paths in the stripping stage, Li in the interior is hard to be reutilized.

The ionic pathways in the mixed conducting framework are provided by the polymer and ceramic, or their composites. However, these SSEs cannot function as effectively as their flowing non-aqueous electrolyte counterparts do. The latter are easily accessible to the interior pores of the conductive matrix. It requires additional wisdom to incorporate solid-state ionic matrix into the anode framework before cell cycling. Zhang et al. integrated both ion and electron conducting paths into a mixed scaffold.⁹⁹ By incorporating $\text{Li}_{6.4}\text{La}_3\text{Zr}_2\text{Al}_{0.2}\text{O}_{12}$ nanoparticles into 3D carbon nanofibers, a mixed ion- and electron-conducting network was fabricated (Figure 10B). Li-metal anodes with these mixed conducting matrixes can deliver an areal capacity of $16\ \text{mAh cm}^{-2}$, volumetric capacity of $1,600\ \text{mAh cm}^{-3}$, and remain stable over 1,000 hr under a current density of $5\ \text{mA cm}^{-2}$. However, there is a massive number of liquid electrolytes in the cell, and safety concerns in practical applications cannot be ignored.



The 3D solid mixed conducted framework can potentially realize a high-capacity, dendrite-free, and safe Li-metal anode. However, similar to the cathode, it is really hard for the anode to construct an interconnected ionic path by SSE. The ion transport just occurs at the interface for the routine Li-metal anode coated by SSE, which is similar to the ultrathin cathode. However, fabricating 3D interconnected and robust ionic paths inside a Li-metal anode is a great challenge.

The ultimate goal is to construct an all-solid-state LMB with a mixed ion- and electron-conducting framework in the Li-metal anode, dense and flexible electrolyte, and the interfacial layer (Figure 11).

- (1) The mixed conducting framework is constructed by the highly ion-conductive porous SSE and electron-conductive host, which can render the rapid transfer of electrons and ions, suppress dendrite growth, relieve the volume expansion, and increase the electrode utilization.
- (2) The dense and flexible SSE separator is probably a composite of ceramic and polymer, combining the flexibility and high ionic conductivity of both.

- (3) The interface between SSE and Li-metal electrode can be constructed by lithiophilic coating or non-aqueous wetting. This super-structure is an ideal cell pattern, which requires the collaboration of researchers from solid-state physics, surface and interface chemistry, material chemistry, simulation, and engineering.

Some fundamental principles should be considered when designing the super-structure:

- (1) Li-metal matrix. Inside the mixed conducting framework, each site of the matrix should be equipotential to make sure that Li ions plate and strip uniformly inside the framework rather than just occurring on the surface.
- (2) Composite SSE separator. What is the ratio of ionic paths for the ceramic, polymer electrolyte, and their interfaces? How do Li ions transfer in the composite electrolyte?
- (3) Interface modifications between Li-metal anode and SSE. What is the diffusion path in the interface, and how do Li ions diffuse through it? If non-aqueous electrolyte is adopted to wet the interface, what is the reaction between non-aqueous electrolyte SSE and Li metal? What is the existence form for the non-aqueous electrolyte after activations?

When designing the super-structure, novel fabrication methods and engineering technologies are required to synthesize these complicated materials.

- (1) Mixed conducting framework. There are at least three phases in the composites: Li metal, electronically conductive materials, and ionically conductive materials. A wise strategy is to build the mixed conducting matrix at first and then fill Li metal into the matrix. Due to the low melting point (180°C) of Li metal, it is possible to fill molten Li into the matrix by capillarity role or press Li metal into the matrix at a relatively high temperature. There are still many troubles in incorporating the electronic and ionic conductor together by sintering, dipping, or coating.
- (2) Composite SSE separator. It is easy to obtain composite SSE just by mixing them together. However, the conductive ceramic must be interconnected to realize a high apparent conductivity, which is hard to realize by simply mixing. A high compactness of the composite SSE is necessary to avoid the generation of voids. Additionally, the composite SSE separator should be thin and light enough, on the contrary, resulting in a great demand on the SSE flexibility to avoid the fracture.
- (3) Interface modifications between Li-metal anode and SSE. If the lithiophilic coating materials are adopted, which methods can be adopted to facilitate and efficiently coat the lithiophilic materials to the surface of Li metal and SSE, while ensuring the intimate contact between them? Wetting the interface using non-aqueous electrolytes can be facilitated by injecting them into the interlayer between Li metal and SSE separator, while its functional mechanism is still not clear. The electrolyte species and proper usage must be clearly investigated through many trials and errors.

CONCLUSION AND PERSPECTIVE

Batteries incorporating Li-metal anode and SSE are strongly recognized as one of the next-generation energy-storage systems with high energy density, long lifespan, and adequate safety assurance. Extensive progress has been made, such as the high

ionic conductivity of bulk SSE (even higher than $10^{-2} \text{ S cm}^{-1}$ at working temperatures) and dendrite-suppressing ability on Li-metal anode in the non-aqueous electrolyte (even at a large current density of 15 mA cm^{-2}). However, design and fabrication of high-performance SSLMBs, especially at room temperature, still remains a huge challenge because of dendrite growth on Li-metal anode and the large interfacial resistance between Li-metal anode and SSE. The cycling current ($0.01 \sim 3 \text{ mA cm}^{-2}$), capacity ($0.0025 \sim 3 \text{ mAh cm}^{-2}$), and utilization of Li-metal anode ($<95\%$) for SSLMBs at room temperature are still very low. Both surface and interface engineering, including composite electrolyte and lithiophilic coating, have been proven to be successful initial strategies for improving the flexibility of SSE, reducing the interfacial resistance and side reactions.

SSLMBs have their brilliant prospects, though practically there are still many barriers to be overcome. Through further investigations into the science and engineering of Li-metal anode and SSE, rechargeable SSLMBs are quite promising to be applied practically. Several principles can be considered in future research on SSLMBs:

- (1) Surface chemistry. SSLMB is a complicated system, requiring comprehensive understandings on the role of SSE and Li-metal anode, especially operating at a large current and capacity. The surface and interface chemistry and materials evolution at the interface between SSE and Li-metal anode are still unknown yet. The current and capacity adopted so far are usually less than 1.0 mA cm^{-1} and 1.0 mAh cm^{-2} , which is far from the practical requirements of 10.0 mA cm^{-2} and 10.0 mAh cm^{-2} .
- (2) Li-ion diffusion and reduction at the interfaces. Li ions from the bulk electrolyte have to diffuse through the interface and get reduced to Li metal. How do Li ions diffuse through the interfacial layer? Where do Li ions get electrons in the composite anode framework? These are still open questions.
- (3) Stable interface construction. Nanoscience provides more opportunities to effectively regulate the component and structure of the artificial SEI. How should the component and structure of the interfacial film be accurately controlled? Fundamental understanding its chemical mechanisms is critically significant to construct a stable interface.
- (4) Reduction in the interfacial impedance. The solid-solid contact between SSE and Li metal can lead to a large interfacial impedance, while its origin and regulation rules are still not clear. The large impedance can result in fatal effects on the cycling performances of solid-state batteries, especially at large currents or under other external unexpected destroy. Is there any self-healing method to reduce the interfacial impedance efficiently and permanently?
- (5) Compatibility with cathodes. Under abuse conditions, oxide cathode can release oxygen, which can reach the SSE and Li-metal anode and lead to performance deterioration and safety concerns. Clearly understanding the materials evolution and chemical mechanism contributes to controllably releasing the energy and efficiently maintaining the safety assurance when solid-state Li-metal anode matches different cathodes, especially the high-voltage intercalation cathodes.
- (6) Characterizations of the working batteries. The features of SSE, Li-metal anode, and their interfaces at different charge and discharge depths are tightly sealed to researchers. An *in situ* and *operando* characterizations on the working batteries can help researchers obtain the exact evolution process and chemical mechanism of different agents in the solid-state batteries and regulate them accurately and efficiently.

- (7) High-throughput screening. Artificial intelligence and machine learning can greatly improve the efficiency in the materials screening of SSE, interfacial layer, electron-conductive matrix, and other cell components. Combining the high-throughput screening and theoretical chemistry, physical chemistry, and material chemistry can vastly improve the research progress of SSLMBs.
- (8) Cell system construction. A functional energy system includes not only a cathode, electrolyte, and anode but also the battery management systems. Combining chemical information of electrode and electrolyte materials with battery management and model prediction can possibly decouple the relationships between the real-time cell energy density and safety and the charge-discharge depth in a cell unit and package scale, which is the only way for a commercial battery.
- (9) Implant of research paradigm. The theory and mechanism of SSLMBs can be implantable to other energy storage systems. Understanding the relations between solid-state batteries and energy chemistry can boost the development of next-generation energy storage systems.

It is the best of time for the development of energy storage systems, and it is also the dawn for SSLMBs if their practical applications could ever see the light of day. More efforts are required to boost the progress. Theoretical chemistry presents novel insights into the chemical reactions and a possibility to rapidly select the chemical systems; materials chemistry allows researchers to accurately synthesize the electrolyte and electrode materials; energy chemistry can clearly provide the interfacial features between SSE and Li-metal anode. The synergism from chemistry, engineering, energy, materials, mechanics, and battery management sheds new lights in the practical applications of SSLMBs. It is the spring of hope for SSLMBs, and we can have all the premises to overcome the difficulties with persistent researches from multidisciplinary fields.

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AUTHOR CONTRIBUTIONS

Q.Z. proposed the topic of the review. X.C., C.Z., Y.Y., H.L., and Q.Z. discussed and revised the manuscript.

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